Google Trends in Depth

The nuts, the bolts, and the gotchas

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Why a whole session on one data source

You scraped it this morning and modeled with it before lunch. Now the fine print.

- Google Trends is the backbone signal of digital epidemiology, and it is **deceptively messy**. Used naively, it can *hurt* a forecast.
- Two halves, both from our own work:
  - **The gotchas and the fix**: how the data is really generated, and how to preprocess it so it forecasts again.
  - **Completing the picture**: how to find the *right* related queries, and what they reveal about disease progression.

The goal: leave able to spot a bad Google Trends signal, and to fix it.

Roadmap

What Google Trends actually is

The four gotchas

The fix: statistical preprocessing

Completing the picture: related queries

Wrap-up

What Google Trends actually is

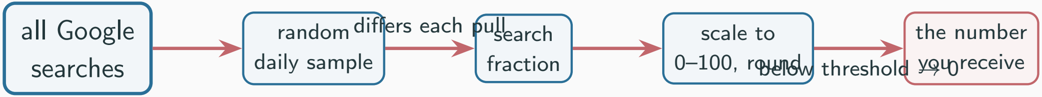
It is not a search count

Google Trends never gives you the number of searches. It gives **relative interest**, and that changes everything.

- Values are scaled **0 to 100** *within* the query set, time window, and geography you asked for. Change any of those and every number moves.
- It is computed from a **random sample** of Google searches, re-drawn each time. Pull the same query twice and you can get different numbers.
- Low volumes are rounded, and below a privacy threshold they are reported as **zero**.

A single raw pull is one noisy, rescaled, partly censored draw. Not the truth.

How a Google Trends value is made



- Every step that makes it privacy-safe and comparable also makes it **unstable**: sampling, rounding, thresholding, rescaling.
- So cleaning it up is a **statistics problem**, not an engineering one.

The four gotchas

Raw Google Trends has four structural problems

From *Restoring the Forecasting Power of Google Trends with Statistical Preprocessing* (Djorno, Yang, et al., *IJF* 2026):

1. **Missing values** (zeros from the privacy threshold),
2. **Sampling variability** (different numbers across downloads),
3. **Noise** (jitter even in non-zero series),
4. **Trend** (magnitudes shift when Google changes its pipeline).

And it is getting worse

Quality has **deteriorated** recently: more zeros and more noise, even for queries that used to be stable. Old code that worked in 2016 can quietly fail now.

Gotcha 1: zeros from the privacy threshold

- Search volumes below a privacy cutoff are reported as **0**, not as a small number.
- A modestly popular term (say "influenza treatment") can come back **mostly zeros**, especially at the state level or off-season.
- Those zeros are not "no interest"; they are **censored** data. Feed them to a model as real values and you teach it nonsense.

Tell-tale sign
A column that is $>$ 50–80% zeros.

Standard move: drop columns with $>$ 80% zeros, or combine terms to lift them above the threshold (later).

Gotcha 2: sampling variability

- The 0–100 series is built from a **fresh random sample** each time Google computes it. Two downloads of the same query, on different days, **disagree**.
- This is not rounding. The underlying draw is different, so historical values you pulled last month may not match what you pull today.

Consequence for you
Your "data" is a random variable. A model fit on one pull is fit on one realization. The cure is to treat it like sampling: **pull repeatedly and average**, so the noise cancels.

Gotchas 3 and 4: noise and trend

Noise

- Even a clean, non-zero series jitters week to week beyond real behavior.
- Low signal-to-noise swamps the actual epidemic signal.

Trend (the sneaky one)

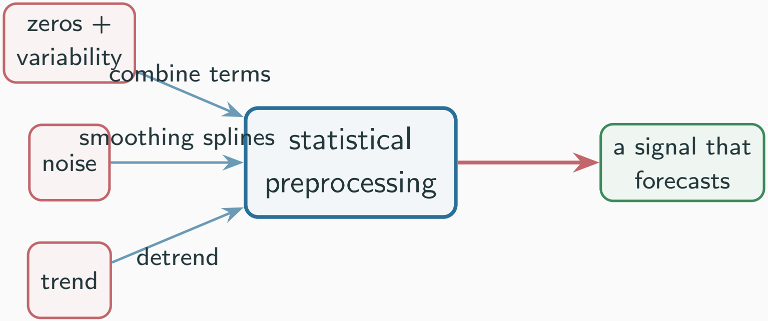
- Google periodically changes its data collection (it even annotates this on the site).
- Magnitudes drift over the years for reasons that have *nothing* to do with searching. A spurious trend the model will happily fit.

Four different problems. They need four different, deliberate fixes.

The fix: statistical preprocessing

Three fixes, and a headline result

The Restoring recipe: **hierarchical clustering, smoothing splines, detrending**.



Validated by forecasting U.S. flu hospitalizations with ARIMAX: raw Google Trends ***hurt****; preprocessed improved accuracy* ***58% nationally, 24% by state****.*

Fix 1: combine related terms (beats zeros and variability)

- Cluster related queries, then **sum** them into one series (Google's Boolean additive query, `term1 + term2 + ...`).
- Combining lifts low-volume terms **above the zero threshold**, so the censoring problem largely disappears.
- Summing many partly-independent draws also **averages out** the sampling noise. One good composite beats many sparse columns.

This is also the bridge to Part B: which terms belong together?

Fix 2 and Fix 3: denoise, then detrend

Denoise (smoothing splines)

- Fit a smooth curve through the jitter; keep the epidemic-scale signal.
- Combined with repeated pulls and averaging, this lifts the signal-to-noise ratio substantially.

Detrend

- Estimate the slow deterministic drift from pipeline changes and **remove it**.
- What remains moves with the disease, not with Google's algorithm updates.

The lesson
A fancier forecasting model did not rescue Google Trends. **Careful preprocessing did.** Clean the input before you blame the model.

Completing the picture: related queries

One term is never the whole story

From our in-progress work with C. Djorno (Djorno, Slavin, Yang, Meyer, Santillana):

- "dengue" or "flu" alone captures a slice of behavior. The full outbreak shows up across a **family** of queries: prevention, symptoms, treatment, complications.
- To see the whole disease, you have to **find the related queries** and use them together, not guess one keyword.

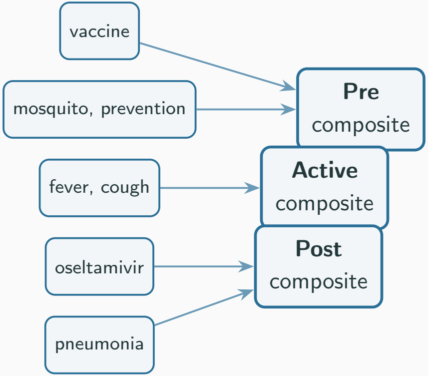
Retrieving related queries is what makes the Google Trends picture complete.

Trick: use topics, not raw strings

- A raw string ("flu") misses spellings, synonyms, and other languages.
- Google Trends **topics** aggregate semantically equivalent searches across spellings and languages into one entity.
- Topics carry **more volume** and **fewer zeros**, which directly helps with Gotcha 1. Prefer topics whenever they exist.

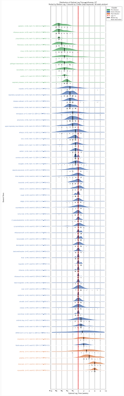
How to build the pool
Seed with the obvious terms, harvest **related and rising** queries, and pull topic IDs (e.g., from Wikidata). Then filter: drop columns $> 80\%$ zeros, and keep terms that actually correlate with the disease at some lag.

Then group them into stage composites



- Cluster terms by **when** they peak relative to cases, then sum each cluster into one composite (Fix 1 again).
- Result: a few clean, near-orthogonal signals, not hundreds of sparse columns.

The payoff: search encodes a three-stage timeline

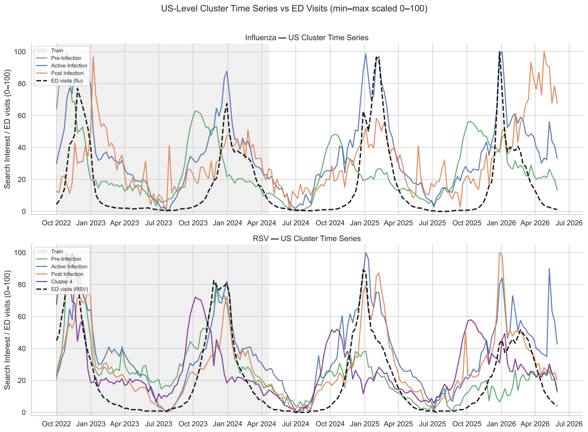


Related queries sort into three stages by their lag to ED visits:

- **Pre** (prevention): leads by **4–7 weeks**,
- **Active** (symptoms, acute care): about **0**,
- **Post** (complications): lags by **3–5 weeks**.

Holds across Google Trends *and* clinician (UpToDate) searches, for both influenza and RSV, in over 90% of states.

The three composites track the wave at different offsets



Pre leads, Active is on-peak, Post trails, all overlaid on ED visits.

Why the stage view matters

- It is not just "search predicts cases." The *category* of search tells you **which stage** of the outbreak you are in.
- Different stages help at different **forecast horizons**: prevention terms for longer lead times, symptom terms for the near term.
- It replicates across a completely different data source (clinician searches), so it is a **real epidemiological structure**, not a Google Trends artifact.

Two honest caveats

Search reflects attention, not only infection (news moves it too), and relationships drift as a disease becomes familiar. Preprocess, validate out of sample, re-check each season.

Wrap-up

Google Trends field guide (screenshot this)

- Never trust a **single raw pull**: pull repeatedly and average.
- Prefer **topics** over raw strings.
- **Combine** related terms; drop columns that are mostly zeros.
- **Denoise** (smoothing) and **detrend** (remove pipeline drift).
- Expect quality to **drift over time**; re-validate old pipelines.
- Always judge on **out-of-sample** forecast error, not in-sample fit.

Raw Google Trends can hurt a model. Preprocessed, it is a genuine early signal.

Looking ahead

- **Next (3:30):** use an agent to scrape more streams (Wikipedia, wastewater) the same careful way.
- **Day 2:** feed these *preprocessed*, stage-grouped search signals into **ARGO**, alongside the disease's own history.

You now know both how to get a Google Trends signal and how to make it trustworthy. That is most of the battle.

Questions?